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**Park et al.**

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(54) **DIRECT CURRENT/DIRECT CURRENT  
CONVERTER AND CONTROL METHOD  
THEREOF**

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(52) **U.S. Cl.**

CPC ..... **H02H 7/1213** (2013.01)

(58) **Field of Classification Search**

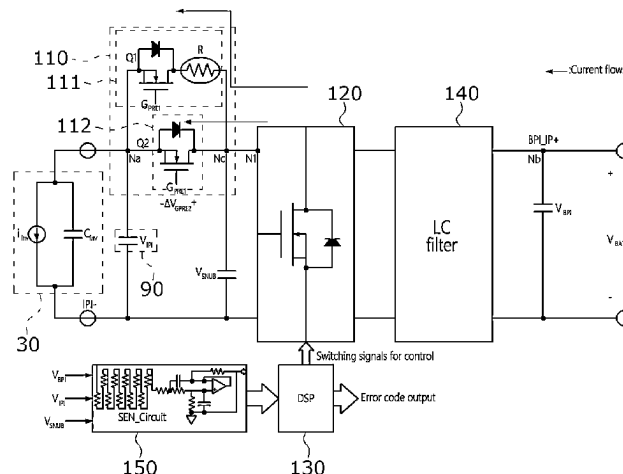
CPC ..... H02H 7/1213; H02M 5/45; H02M 1/32;  
H02M 1/36

See application file for complete search history.

(57) **ABSTRACT**

According to an embodiment, provided is a direct current/  
direct current converter comprising: a first switch unit  
located between a direct current link capacitor and a bridge  
circuit unit; a second switch unit connected in parallel with  
the first switch unit; a sensing unit for sensing a voltage  
across both ends of an inverter and a voltage across both  
ends of a second switch; and a control unit which controls  
an on-off operation of the first switch unit and the second  
switch unit, and determines whether a short circuit or a  
disconnection occurs by using the voltage across both ends  
of the inverter and the voltage across both ends of the second  
switch unit, wherein the control unit determines whether a  
short circuit or a disconnection occurs by using a voltage  
across both ends of the inverter and a voltage across both  
ends of the second switch unit, which are measured when the  
first switch unit is in an on state and the second switch unit  
is in an off state, and determines whether a disconnection  
occurs by using a voltage across both ends of the inverter,

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which is measured when the second switch unit is in an on state and the first switch unit is in an off state.

**20 Claims, 7 Drawing Sheets**

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FIG. 1

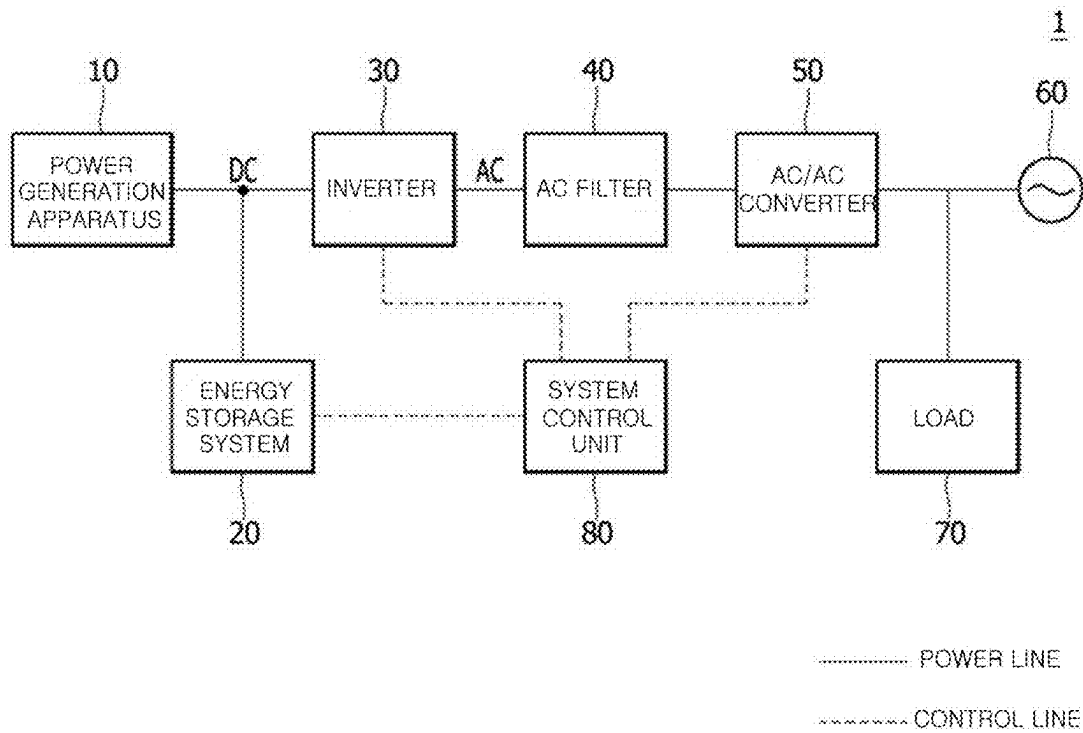


FIG. 2

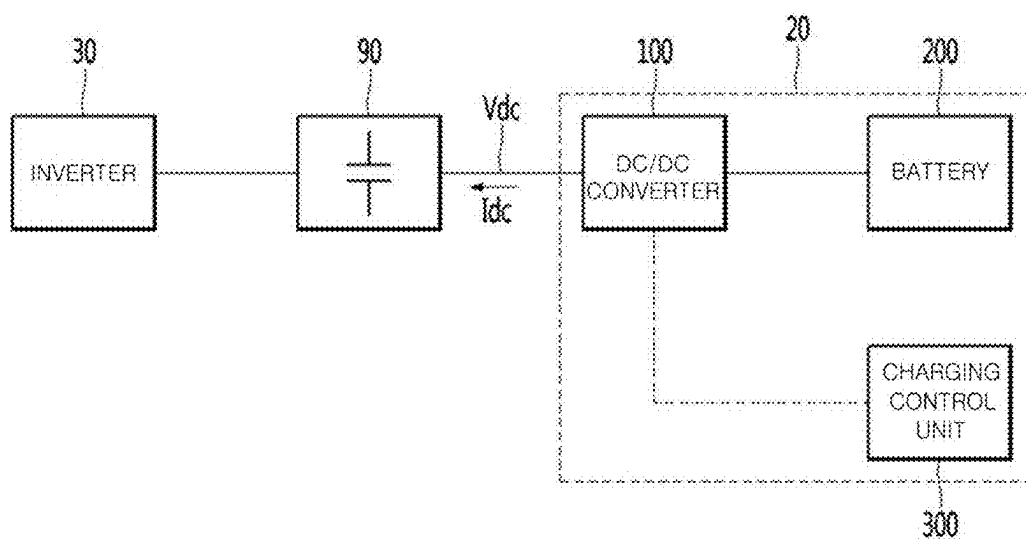


FIG. 3

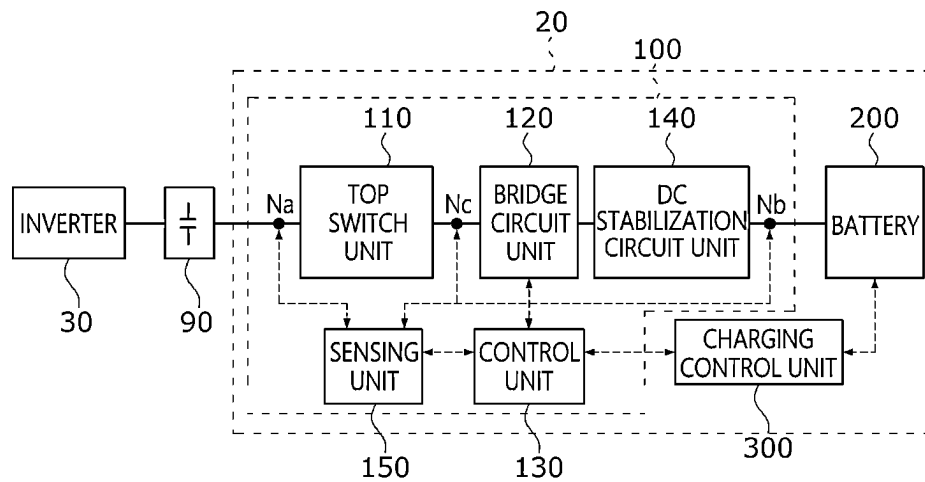


FIG. 4

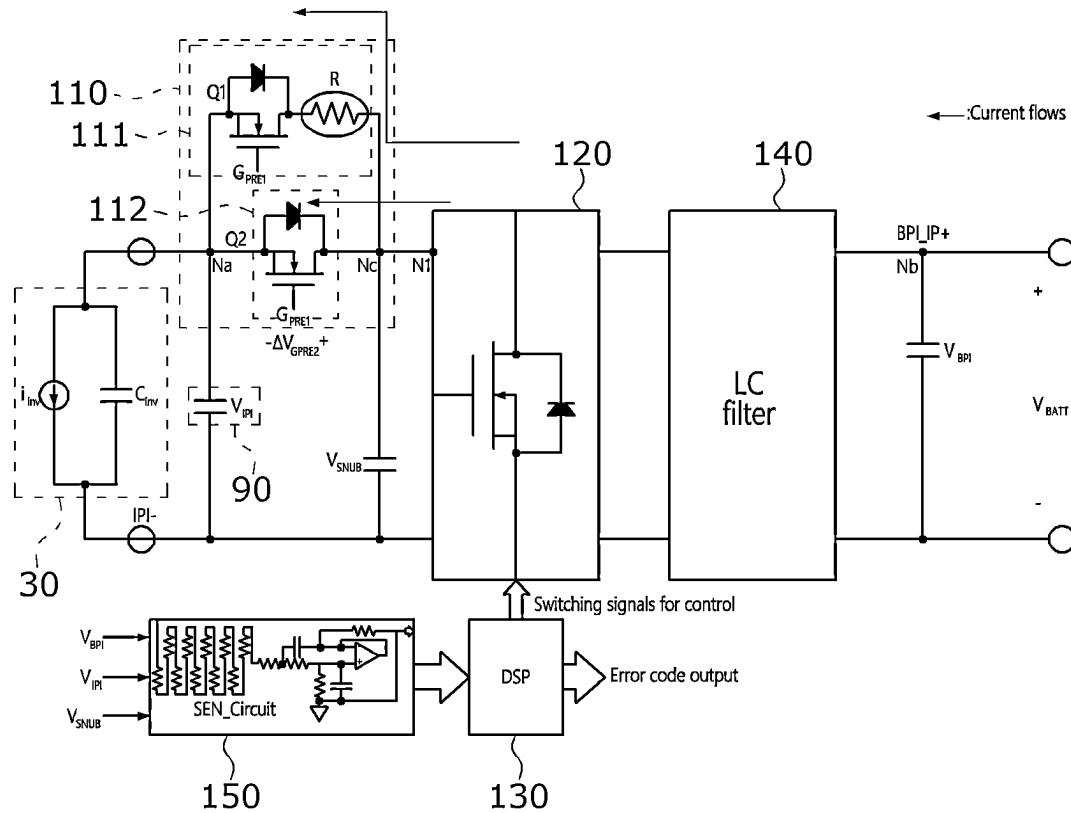
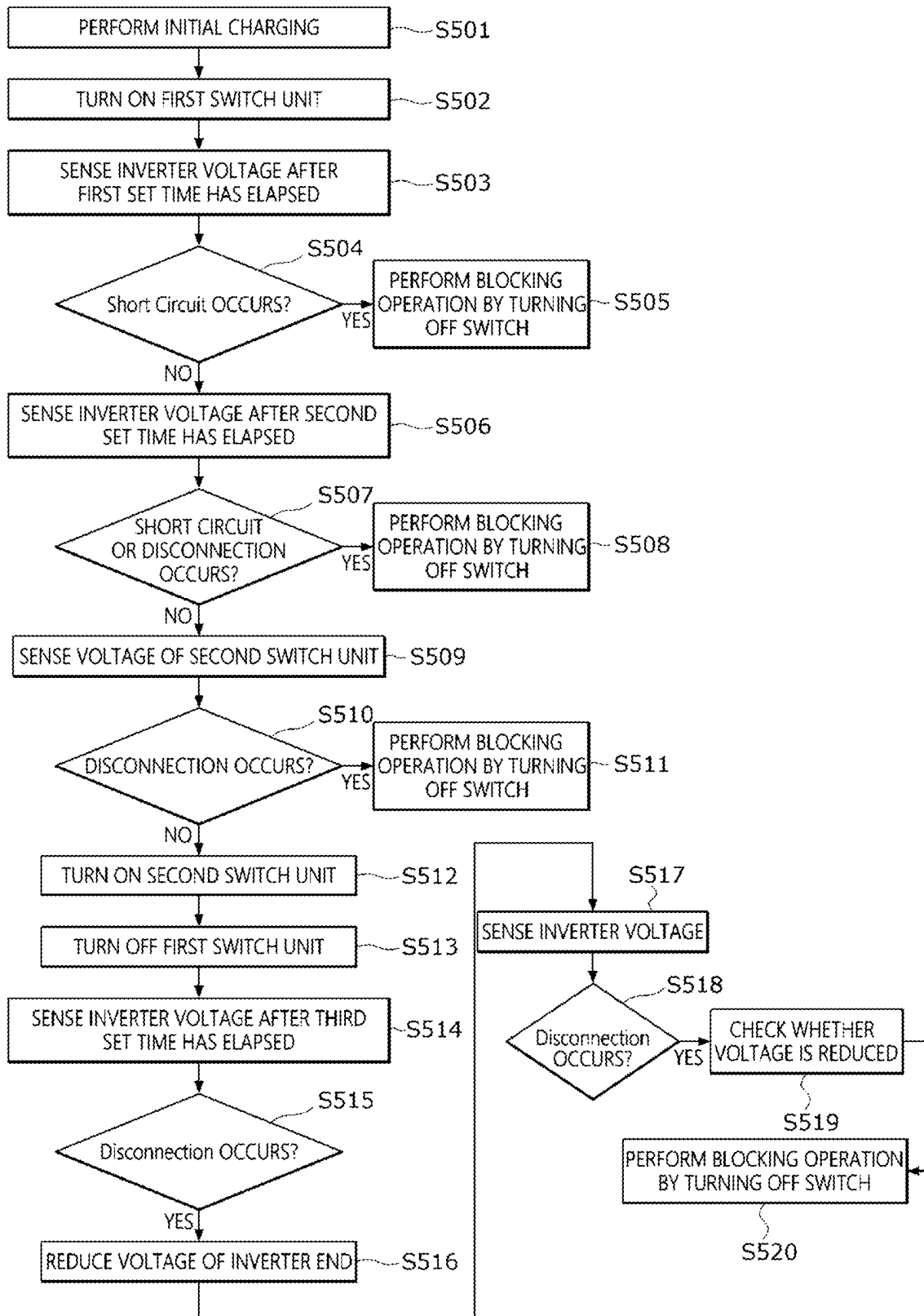


FIG. 5



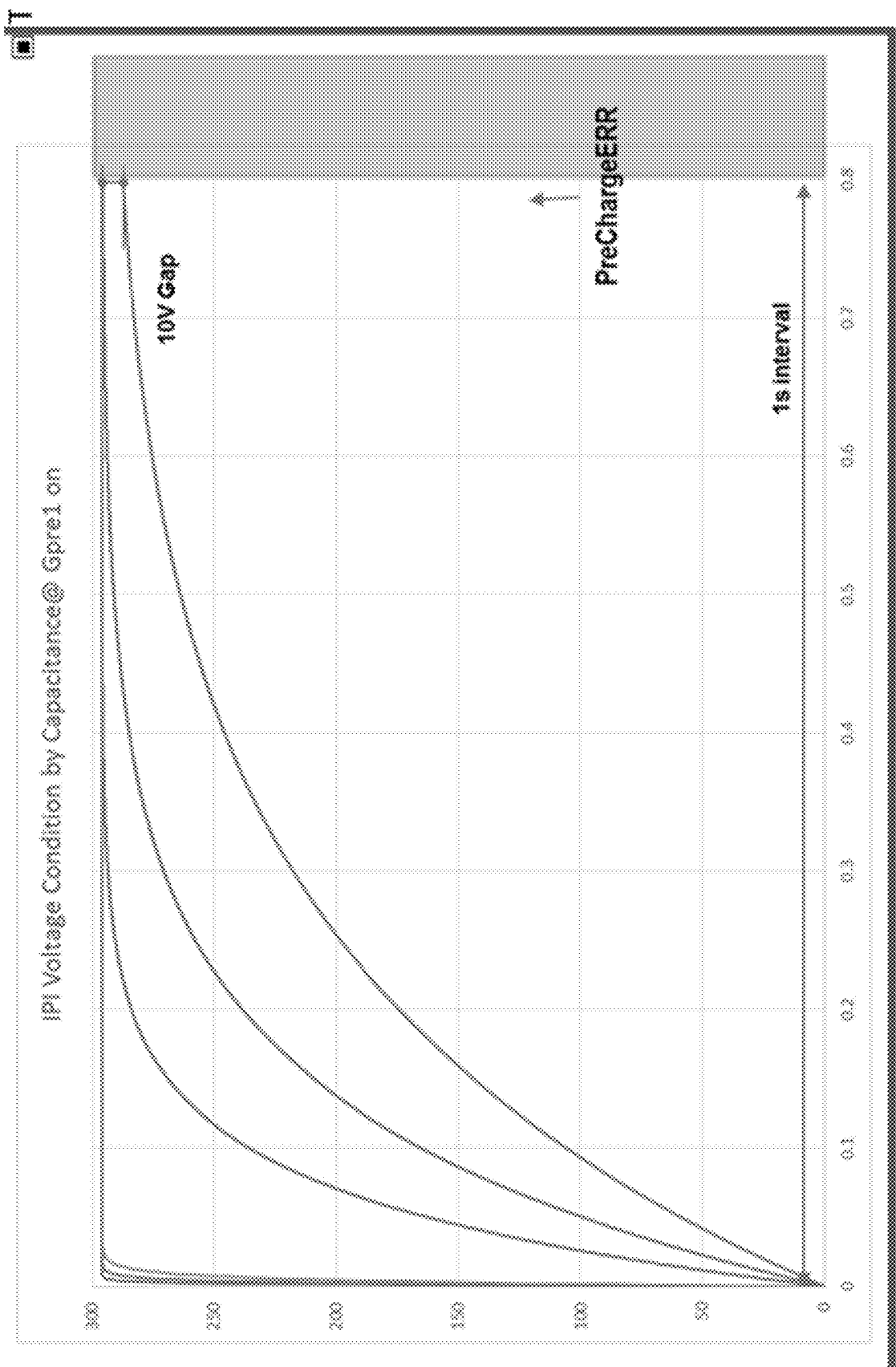


FIG. 6A

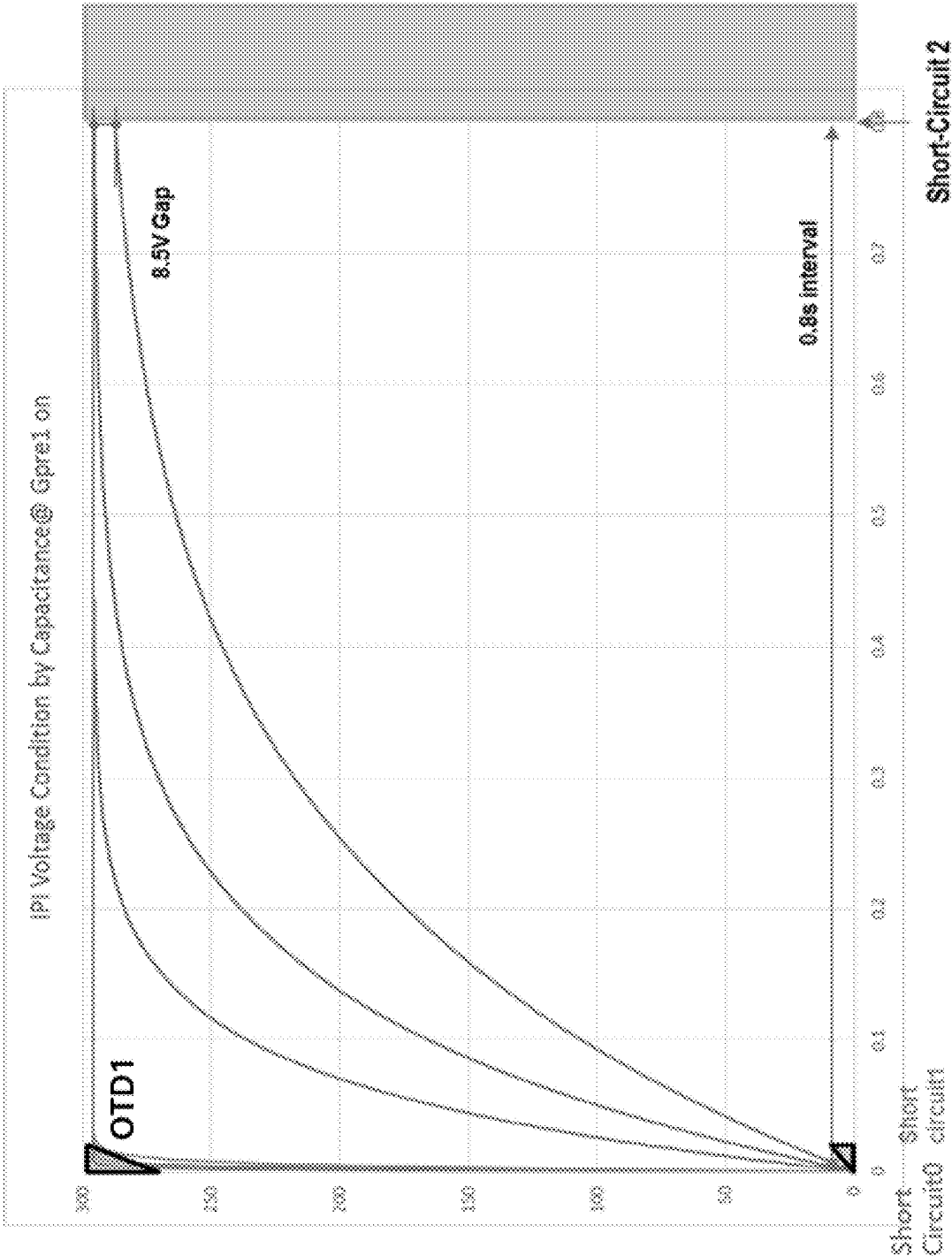


FIG. 6B

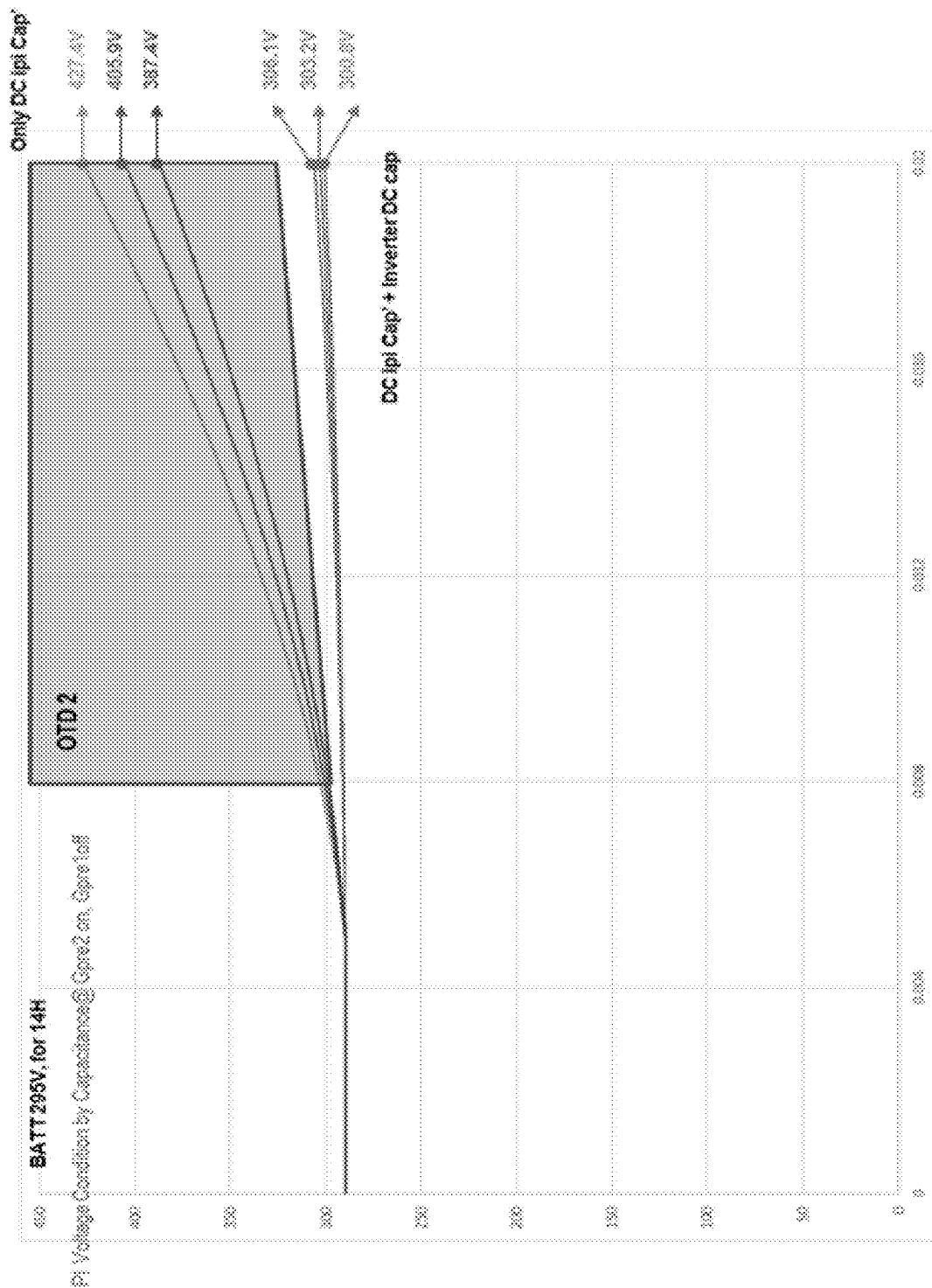
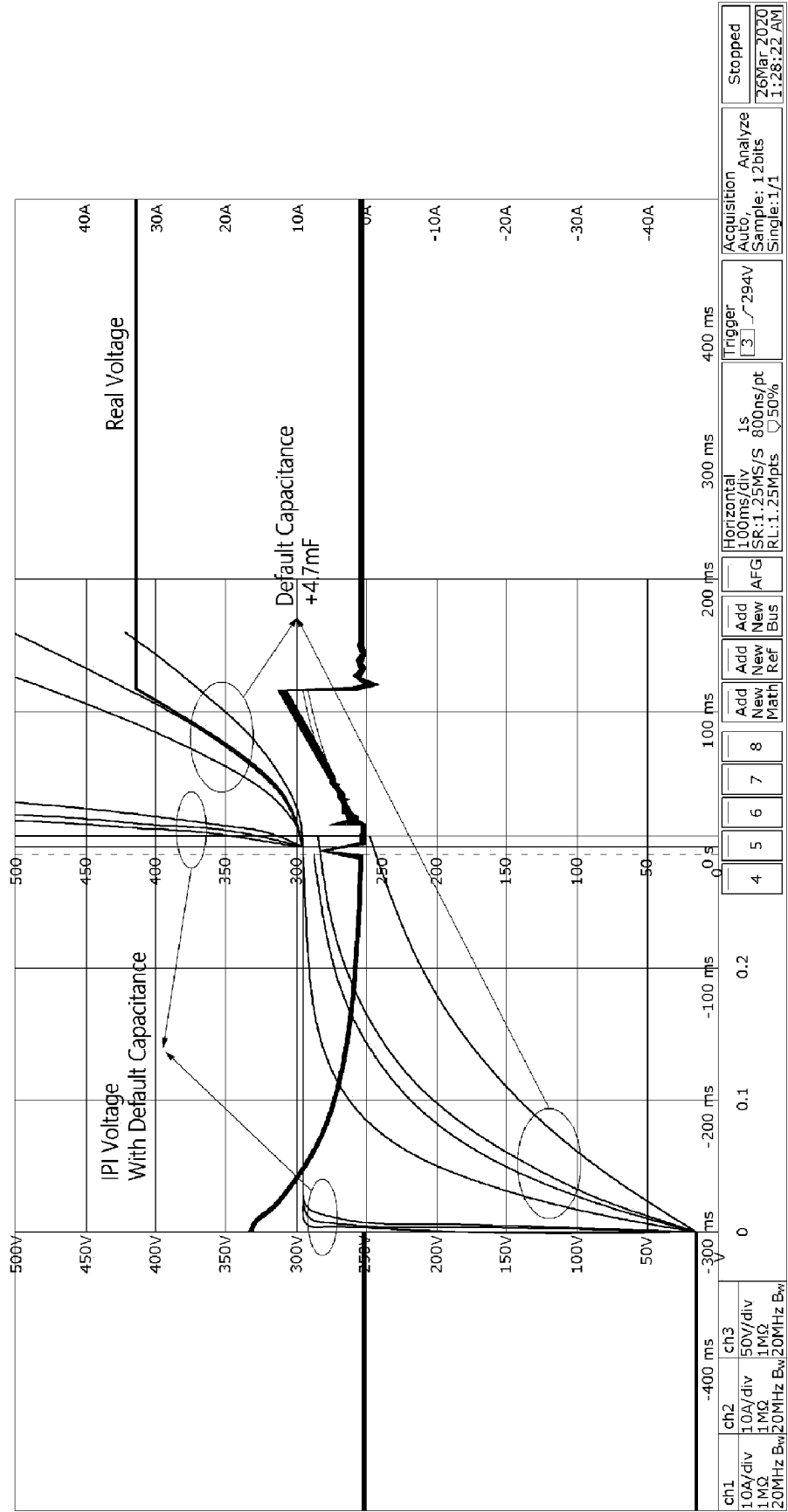


FIG. 6C

FIG. 6D



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# **DIRECT CURRENT/DIRECT CURRENT CONVERTER AND CONTROL METHOD THEREOF**

## **CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is the U.S. national stage application of International Patent Application No. PCT/KR2021/011655, filed Aug. 31, 2021, which claims the benefit under 35 U.S.C. § 119 of Korean Application No. 10-2020-0160980, filed Nov. 26, 2020, the disclosures of each of which are incorporated herein by reference in their entirety.

## **TECHNICAL FIELD**

One embodiment of the present invention relates to a direct current/direct current (DC/DC) converter and a control method thereof.

## **BACKGROUND ART**

Electric energy is easily converted and transmitted, and thus is widely used. In order to efficiently use this electric energy, an energy storage system (ESS) is used. The energy storage system receives power and charges the power in a battery. Further, when power is required, the energy storage system supplies the power by discharging the power charged in the battery. Accordingly, the energy storage system can flexibly supply power.

Specifically, a power supply system operates as follows when including the energy storage system. The energy storage system discharges electric energy stored in the battery when a load or system is in an overload. Further, when the load or system is in a light load, the energy storage system receives power from a power generation apparatus or system and charges the power in the battery.

In addition, when the energy storage system is present independently of the power supply system, the energy storage system receives idle power from an external power source and charges the idle power in the battery. In addition, when a system or load is in an overload, the energy storage system supplies the power by discharging the power charged in the battery.

Meanwhile, a direct current/direct current (DC-DC) converter applied to the energy storage system operates in a state in which detection of a short circuit and a disconnection is not performed. The DC-DC converter can cause damage to components and damage to workers when operating in a short-circuited or disconnected state.

## **DISCLOSURE**

### **Technical Problem**

The present invention is directed to providing a direct current/direct current (DC/DC) converter capable of performing detection of a short circuit and a disconnection, and a control method thereof.

### **Technical Solution**

According to an embodiment, a direct current/direct current (DC/DC) converter control method includes: turning on a first switch unit located between a direct current link capacitor and a bridge circuit unit; determining whether a short circuit (short) occurs after a first set time has elapsed;

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determining whether the short circuit and a disconnection (open) occur after a second set time has elapsed; measuring a voltage across both ends of a second switch unit connected in parallel with the first switch unit; comparing the voltage across both ends of the second switch unit and a preset voltage to determine whether the short circuit occurs; turning on the second switch unit; turning off the first switch unit, and determining whether the disconnection occurs after a third set time has elapsed from a time when the first switch unit is turned off.

The determining of whether the short circuit (short) occurs after the first set time has elapsed may include comparing a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of an inverter and determining whether the short circuit occurs.

The determining of whether the short circuit and the disconnection (open) occur after the second set time has elapsed may include comparing a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of an inverter and determining whether the short circuit and the disconnection occur.

The determining of whether the disconnection occurs after the third set time has elapsed may include comparing a preset third reference voltage value and a preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of an inverter and determining whether the short circuit and the disconnection occur.

The comparing of the voltage across both ends of the second switch unit and the preset voltage to determine whether the short circuit occurs may include determining that the short circuit occurs when the voltage across both ends of the second switch unit is maintained under the preset voltage for a preset time.

The DC/DC converter control method may further include outputting at least one of a control command for stopping an operation of a converter and a control command for reducing a voltage of an inverter end when the short circuit or the disconnection occurs.

According to an embodiment, a DC/DC converter includes: a first switch unit located between a direct current link capacitor and a bridge circuit unit, a second switch unit connected in parallel with the first switch unit; a sensing unit configured to sense a voltage across both ends of an inverter and a voltage across both ends of a second switch; and a control unit configured to control an on-off operation of the first switch unit and the second switch unit, and determine whether a short circuit and a disconnection occur using the voltage across both ends of the inverter and the voltage across both ends of the second switch unit, wherein the control unit determines whether the short circuit and the disconnection occur using the voltage across both ends of the inverter and the voltage across both ends of the second switch unit, which are measured when the first switch unit is turned on and the second switch unit is turned off, and determines whether the disconnection occurs using the voltage across both ends of the inverter, which is measured when the second switch unit is turned on and the first switch unit is turned off.

The first switch unit may include a resistor element and a switching element connected to the resistor element in series.

The control unit may sequentially perform arithmetic operations of determining whether a short circuit (short) occurs after a first set time has elapsed after turning on the first switch unit, determining whether the short circuit and a disconnection (open) occur after a second set time has elapsed, and comparing the voltage across both ends of the second switch unit and a preset voltage to determine whether the short circuit occurs, and may determine whether the disconnection occurs after a third set time has elapsed after turning on the second switch unit, and turning off the first switch unit.

The control unit may output at least one of a control command for stopping an operation of a converter and a control command for reducing a voltage of an inverter end when the short circuit or the disconnection occurs.

The control unit may compare a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of the inverter after a first set time has elapsed to determine whether the short circuit occurs.

The control unit may compare a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of the inverter after the second set time has elapsed to determine whether the short circuit and the disconnection occur.

The control unit may compare a preset third reference voltage value and a preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of the inverter after the third set time has elapsed to determine whether the short circuit and the disconnection occur.

#### Advantageous Effects

According to the present invention, in a DC/DC converter and a control method thereof, it is possible to perform detection of a short circuit and a disconnection of the DC/DC converter.

Further, breakage of a circuit and a personal injury can be inhibited by stopping an operation and reducing a voltage of an inverter end when the short circuit or the disconnection is detected.

In addition, precise short circuit detection and disconnection detection can be performed regardless of a change in resistance value according to a temperature of a resistor for inhibiting overcurrent.

In addition, stability can be secured by causing a fault when connecting an inverter over an operation range previously agreed with an inverter company to inhibit an operation.

#### DESCRIPTION OF DRAWINGS

FIG. 1 is a view for describing a schematic configuration of a power supply system according to an embodiment.

FIG. 2 is a view for describing an energy storage system according to the embodiment.

FIG. 3 is a view for describing a direct current/direct current (DC/DC) converter according to the embodiment.

FIG. 4 is a circuit diagram of the DC/DC converter according to the embodiment.

FIG. 5 is a flow chart of an operation of the DC/DC converter according to the embodiment.

FIGS. 6A-6D are views of a result of testing detection performance of a short circuit and a disconnection of the DC/DC converter according to the embodiment.

#### MODES OF THE INVENTION

Hereinafter, preferable embodiments of the present invention will be described in detail with reference to the accompanying drawings.

However, the technical spirit of the present invention is not limited to some embodiments which will be described and may be embodied in various forms, and one or more elements in the embodiments may be selectively combined and replaced to be used within the scope of the technical spirit of the present invention.

Further, terms used in the embodiments of the present invention (including technical and scientific terms), may be interpreted with meanings that are generally understood by those skilled in the art unless particularly defined and described, and terms which are generally used, such as terms defined in a dictionary, may be understood in consideration of their contextual meanings in the related art.

In addition, terms used in the description are provided not to limit the present invention but to describe the embodiments.

In the specification, the singular form may also include the plural form unless the context clearly indicates otherwise and may include one or more of all possible combinations of A, B, and C when disclosed as at least one (or one or more) of "A, B, and C."

In addition, terms such as first, second, A, B, (a), (b), and the like may be used to describe elements of the embodiments of the present invention.

These terms are only provided to distinguish the elements from other elements, and the essence, sequence, order, or the like of the elements are not limited by the terms.

Further, when particular elements are disclosed as being "connected," "coupled," or "linked" to other elements, the elements may include not only a case of being directly connected, coupled, or linked to other elements but also a case of being connected, coupled, or linked to other elements by elements between the elements and other elements.

In addition, when one element is disclosed as being formed "on or under" another element, the term "on or under" includes both a case in which the two elements are in direct contact with each other and a case in which at least another element is disposed between the two elements (indirect contact). Further, when the term "on or under" is expressed, a meaning of not only an upward direction but also a downward direction may be included based on one element.

Hereinafter, embodiments will be described in detail with reference to the accompanying drawings, and the same reference numerals are granted to the same or corresponding components regardless of the drawing numerals and overlapping descriptions thereof will be omitted.

FIG. 1 is a view for describing a schematic configuration of a power supply system according to an embodiment. Referring to FIG. 1, a power supply system 1 according to the embodiment includes a power generation apparatus 10, an energy storage system 20, an inverter 30, an alternating current (AC) filter 40, an alternating current/alternating current (AC/AC) converter 50, a system 60, a system control unit 80, and a load 70.

The power generation apparatus 10 may produce electric energy. When the power generation apparatus 10 is a photovoltaic power generation system, the power generation apparatus 10 may be a solar cell array. The solar cell array is a combination of a plurality of solar cell modules. The solar cell module may be a device which converts solar energy to electric energy to generate predetermined voltage

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and current by connecting a plurality of solar cells in series or parallel. Accordingly, the solar cell array may absorb solar energy and convert the solar energy to the electric energy.

Further, when the power generation apparatus **10** is a wind power generation system, the power generation apparatus **10** may be a fan which converts wind energy to electric energy.

Meanwhile, the power generation apparatus **10** is not limited thereto, and may be configured as a tidal power generation system in addition to the photovoltaic power generation system and the wind power generation system. However, this is only an example, and the power generation apparatus **10** is not limited to the above-mentioned types, and may include all power generation systems which generate electric energy using new renewable energy such as solar heat, geothermal heat, or the like.

Further, the power supply system **1** may supply power only through the energy storage system **20** without the power generation apparatus **10**.

In this case, the power supply system **1** may not include the power generation apparatus **10**.

The inverter **30** may convert direct current (DC) power to alternating current (AC) power. More specifically, the DC power supplied by the power generation apparatus **10** or the DC power discharged by the energy storage system **20** may be converted to the AC power.

The AC filter **40** may filter a noise of the power converted to the AC power. Further, the AC filter **40** may be omitted according to the embodiment.

The AC/AC converter **50** may convert a magnitude of a voltage of the noise-filtered AC power so that the AC power may be supplied to the system **60** or the load **70**, and may supply the converted AC power to the system **60** or the load **70**. Further, the AC/AC converter **50** may be omitted according to the embodiment.

The system **60** is a system in which many power plants, substations, transmission and distribution lines, and loads are integrated and thus generation and use of electric power are performed.

The load **70** may receive electric energy from a power generation system such as the power generation apparatus **10** or the like or the energy storage system **20** to consume power.

The energy storage system (ESS) **20** may be charged by receiving electric energy from the power generation apparatus **10**, and may discharge the charged electric energy according to a power supply and demand condition of the system **60** or the load **70**. More specifically, when the system **60** or the load **70** is in a light load, the energy storage system **20** may receive idle power from the power generation apparatus **10** to perform charging. When the system **60** or the load **70** is in an overload, the energy storage system **20** may discharge the charged power to supply the power to the system **60** or the load **70**. Further, the energy storage system **20** may be connected between the power generation apparatus **10** and the inverter **30** to be electrically connected to the power generation apparatus **10** and electrically connected to the inverter **30**.

The system control unit **80** may control operations of the energy storage system **20**, the inverter **30**, and the AC/AC converter **50**. More specifically, the system control unit **80** may control charging and discharging of the energy storage system **20**. When the system **60** or the load **70** is in an overload, the system control unit **80** may control the energy storage system **20** to supply power and transfer the power to the system **60** or the load **70**. When the system **60** or the load **70** is in a light load, the system control unit **80** may control

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an external power source or the power generation apparatus **10** to supply power and transmit the power to the energy storage system **20**.

FIG. **2** is a view for describing the energy storage system according to the embodiment.

Referring to FIG. **2**, the energy storage system **20** according to the embodiment may include a direct current/direct current (DC/DC) converter **100**, a battery **200**, and a charging control unit **300**. The energy storage system **20** may be connected to the inverter **30** through a direct current link capacitor **90**. That is, the direct current link capacitor **90** may be disposed between the energy storage system **20** and the inverter **30**. Accordingly, the energy storage system **20** may receive a direct current (DC) voltage  $V_{dc}$  of the direct current link capacitor **90** in a charging mode, and provide the DC voltage  $V_{dc}$  to the direct current link capacitor **90** in a discharging mode.

The battery **200** may receive charging power from the DC/DC converter **100** in the charging mode, and perform a charging operation using the received power. Further, the battery **200** may output previously stored power to the DC/DC converter **100** in the discharge mode. In addition, the battery **200** may include a plurality of battery cells to perform charging and discharging operations.

The charging control unit **300** may include a battery management system (BMS). The charging control unit **300** may provide battery state information on a state of the battery **200** to the system control unit **80**. For example, the charging control unit **300** may monitor at least one or more of a voltage, a current, a temperature, a remaining power amount, and a charging state of the battery **200**, and transmit the monitored state information of the battery **200** to the system control unit **80**. Further, the charging control unit **300** may make the plurality of battery cells maintain an appropriate voltage while being charged or discharged. In addition, the charging control unit **300** may operate based on a control signal from the system control unit **80**. In addition, the charging control unit **300** may control the DC/DC converter **100** according to the monitored state information of the battery **200**. In addition, the charging control unit **300** may control the DC/DC converter **100** according to the charging mode or discharging mode. More specifically, the charging control unit **300** may provide a charging control signal or discharging control signal for controlling the DC/DC converter **100** to a converter control unit of the DC/DC converter **100**, and the converter control unit of the DC/DC converter **100** may provide a pulse width modulation (PWM) signal to a switch of the DC/DC converter **100** based on the charging control signal or discharging control signal. Further, the charging control unit **300** may control the DC/DC converter **100** to initially charge the direct current link capacitor **90** in the discharge mode of the battery **200**. That is, the charging control unit **300** may provide an initial charging control signal for controlling the DC/DC converter **100** to the converter control unit of the DC/DC converter **100**, and the converter control unit of the DC/DC converter **100** may provide an initial charging switch signal to the switch of the DC/DC converter **100** based on the initial charging control signal. Further, the charging control unit **300** may control the DC/DC converter **100** to improve power conversion efficiency of the DC/DC converter **100**. More specifically, the charging control unit **300** may provide a power conversion efficiency control signal capable of improving the power conversion efficiency of the DC/DC converter **100** to the converter control unit of the DC/DC converter **100**, and the converter control unit of the DC/DC

converter **100** may provide the PWM signal to the switch of the DC/DC converter **100** based on the power conversion efficiency control signal.

The DC/DC converter **100** may convert a magnitude of the DC power received by the energy storage system **20** in the charging mode or supplied from the energy storage system **20** in the discharging mode. More specifically, the DC/DC converter **100** may convert the DC power provided to the direct current link capacitor **90** from the power generation apparatus **10** or the inverter **30** to a voltage magnitude for charging the battery **200** and provide the DC power to the battery **200**. Further, the DC/DC converter **100** may convert the DC power provided from the battery **200** to a voltage magnitude usable by the inverter **30** and provide the DC power to the direct current link capacitor **90**.

FIG. **3** is a view for describing the DC/DC converter according to the embodiment.

Referring to FIG. **3**, the DC/DC converter **100** may convert a magnitude of the DC power received by the energy storage system **20** in the charging mode or supplied from the energy storage system **20** in the discharging mode. That is, the DC/DC converter **100** may be a bidirectional DC/DC converter. More specifically, the DC/DC converter **100** may convert the DC power provided to the direct current link capacitor **90** from the power generation apparatus **10** or the inverter **30** to a voltage magnitude for charging the battery **200** and provide the DC power to the battery **200**. Further, the DC/DC converter **100** may convert the DC power provided from the battery **200** to a voltage magnitude usable by the inverter **30** and provide the DC power to the direct current link capacitor **90**. In addition, the DC/DC converter **100** may operate in the charging mode, a standby mode, and the discharging mode based on a voltage provided from the direct current link capacitor **90**. That is, the DC/DC converter **100** may monitor the voltage provided from the direct current link capacitor **90** even when a control signal is not provided by the charging control unit **300**, determine whether to operate in the charging mode, the standby mode, and the discharging mode, and operate.

The DC/DC converter **100** may include a top switch unit **110**, a bridge circuit unit **120**, a control unit **130**, a direct current stabilization circuit unit **140**, and a sensing unit **150**.

The control unit **130** may control the bridge circuit unit **120**. For example, the control unit **130** may generate a PWM signal based on the control signal provided from the charging control unit **300** and provide the PWM signal to the bridge circuit unit **120** including a switch. As another example, the control unit **130** may determine an operation mode and reference power according to the magnitude of the voltage provided from the direct current link capacitor **90**. Further, the control unit **130** may generate a PWM signal based on the determined reference power and provide the PWM signal to the bridge circuit unit **120** including the switch.

The top switch unit **110** may inhibit electrical overstress (EOS) or overcurrent from being introduced into or discharged from the energy storage system **20**. The top switch unit **110** may be disposed between a first end Na to which the direct current link capacitor **90** is connected and the bridge circuit unit **120**. Further, the top switch unit **110** may include a circuit breaker (not shown). In this case, the top switch unit **110** may open between the first end Na and the bridge circuit unit **120** when EOS or overcurrent is introduced into the energy storage system **20**. Accordingly, the top switch unit **110** may block input/output of an external current to/from the energy storage system **20**.

The bridge circuit unit **120** may be disposed between the top switch unit **110** and the direct current stabilization circuit unit **140** and electrically connected to each component. The bridge circuit unit **120** may drop the DC voltage of the DC power input from the top switch unit **110** in a step-down mode and output the DC voltage to the direct current stabilization circuit unit **140**. Further, the bridge circuit unit **120** may raise the DC voltage of the DC power input from the direct current stabilization circuit unit **140** in a step-up mode and output the DC voltage to the top switch unit **110**. The bridge circuit unit **120** may include one or more switches. For example, the bridge circuit unit **120** may be an insulated full bridge circuit. As another example, the bridge circuit unit **120** may be an uninsulated full bridge circuit. The present invention is not limited thereto, and the bridge circuit unit **120** may be configured as a half bridge circuit. The bridge circuit unit **120** may operate based on the PWM signal of the control unit **130**.

The direct current stabilization circuit unit **140** may operate to raise the DC voltage in the step-up mode of the bridge circuit unit **120** and drop the DC voltage in the step-down mode. Further, the direct current stabilization circuit unit **140** may be an LC filter. The direct current stabilization circuit unit **140** may be connected to a second end Nb.

The sensing unit **150** may sense the voltage and current of the first end Na and provide the sensed voltage and current to the control unit **130**. The voltage and current of the first end Na may be a DC voltage and a DC current provided by the direct current link capacitor **90**.

Further, the sensing unit **150** may sense the voltage and current of the second end Nb and provide the sensed voltage and current to the control unit **130**. The voltage and current of the second end Nb may be a DC voltage and a DC current provided to the battery **200** by the DC/DC converter.

Further, the sensing unit **150** may sense the voltage and current of a third end Nc and provide the sensed voltage and current to the control unit **130**. The voltage and current of the third end Nc may be the voltage and current at a contact point of the top switch unit **110** and the bridge circuit unit **120**.

The sensing unit **150** may be controlled by the control unit **130**. Accordingly, in another embodiment, an operation mode of charging or discharging of the battery may be quickly determined. Further, in still another embodiment, a separate communication line and communication unit for droop control when the battery is charged or discharged are not required. In addition, in yet another embodiment, the quick droop control may be performed when the battery is charged or discharged.

FIG. **4** is a circuit diagram of the DC/DC converter according to the embodiment.

Referring to FIG. **4**, the control unit **130** of the DC/DC converter **100** according to the embodiment may generate a PWM signal based on a control signal provided from the charging control unit **300**, and provide the PWM signal to the top switch unit **110** including the switch or the bridge circuit unit **120**.

In FIG. **4**, arrows indicate moving paths of the currents.

The top switch unit **110** may include a first switch unit **111** and a second switch unit **112**. The second switch unit **112** may include a second switch element Q2 disposed between one end of the direct current link capacitor **90** and a first node N1. The first switch unit **111** may be connected in parallel with the second switch unit **112**. The first switch unit **111** may include a first switch element Q1 and a resistor R. In the first switch element Q1, one side may be connected to

one end of the direct current link capacitor **90** and the other side may be connected to one side of the resistor **R**. In the resistor **R**, one side may be connected to the other side of the first switch element **Q1** and the other side may be connected to the bridge circuit unit **120**. The resistor **R** may make a current of a lower level than a current flowing through the first switch unit **111** flow through the first switch unit **111**.

The bridge circuit unit **120** may be disposed between the top switch unit **110** and the direct current stabilization circuit unit **140**.

The direct current stabilization circuit unit **140** may connect between the bridge circuit unit **120** and the battery **200**.

The sensing unit **150** may sense the voltage and current of the first end **Na** and provide the sensed voltage and current to the control unit **130**. The voltage of the first end **Na** may be the DC voltage and the DC current provided by the direct current link capacitor **90**.

Further, the sensing unit **150** may sense the voltage and current of the second end **Nb** and provide the sensed voltage and current to the control unit **130**. The voltage and current of the second end **Nb** may be the DC voltage and the DC current provided to the battery **200** by the DC/DC converter.

Further, the sensing unit **150** may sense the voltage and current of the third end **Nc** and provide the sensed voltage and current to the control unit **130**. The voltage of the third end **Nc** may be the voltage and current at a contact point of the top switch unit **110** and the bridge circuit unit **120**.

The sensing unit **150** may be controlled by the control unit **130**.

When the DC/DC converter **100** according to the embodiment operates in the step-down mode which is a charging mode, or operates in the step-up mode operation which is a discharging mode, the second switch unit **112** may be turned on/off and the first switch unit **111** may maintain an off state. Further, in an initial charging mode of the DC/DC converter **100** before starting the discharging mode, the second switch unit **112** may maintain an off state and the first switch unit **111** may be turned on/off.

Accordingly, the energy storage system according to the embodiment may initially charge the direct current link capacitor without a separate configuration before operating in the discharge mode. Further, since the energy storage system according to the embodiment gradually increases the DC current provided to the direct current link capacitor, an initial charging speed may be fast and the battery discharging operation may be quickly performed.

According to the embodiment, the control unit **130** may control an on-off operation of the first switch unit **111** and second switch unit **112**, and determine whether a short circuit and a disconnection of the inverter occur using a voltage across both ends of the inverter and a voltage across both ends of the second switch unit **112**.

Further, the control unit **130** may output at least one of a control command for stopping an operation of the DC/DC converter **100** and a control command for reducing a voltage of an inverter end when the short circuit or disconnection occurs. For example, the control unit **130** may block introduction of the current by operating the circuit breaker of the top switch unit **110** when the short circuit or disconnection occurs.

The control unit may determine whether the short circuit and the disconnection occur using the voltage across both ends of the inverter and the voltage across both ends of the second switch unit **112**, which are measured when the first switch unit **111** is turned on and the second switch unit **112** is turned off.

The control unit **130** may operate the first switch unit **111** in an on state, and determine whether a short circuit (short) occurs after a first set time has elapsed. Specifically, the control unit **130** may determine whether the short circuit occurs by comparing a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of the inverter after the first set time has elapsed. The first set time may be set to a predetermined time after the first switch unit **111** is converted to the on state. For example, the first set time may be set to 5 ms, but is not limited thereto. The control unit **130** may calculate the first inverter voltage value and the first inverter current value using the voltage and current of the first end **Na** sensed by the sensing unit **150**, and determine whether the short circuit occurs by comparing the first inverter voltage value and the first inverter current value with the preset first reference voltage value and the preset first reference current value.

Further, the control unit **130** may determine whether the short circuit and the disconnection (open) occur after a second set time has elapsed. Specifically, the control unit **130** may determine whether the short circuit and the disconnection occur by comparing a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of the inverter after the second set time has elapsed. The second set time may be set to be longer than the first set time. For example, the second set time may be set to 10.0 ms, but is not limited thereto. The control unit **130** may calculate the second inverter voltage value and the second inverter current value using the voltage and the current of the first end **Na** sensed by the sensing unit **150**, and may determine whether the short circuit and the disconnection occur by comparing the second inverter voltage value and the second inverter current value with the preset second reference voltage value and the preset second reference current value.

Further, the control unit **130** may determine whether the short circuit occurs by comparing the voltage across both ends of the second switch unit **112** and a preset voltage. Specifically, the control unit **130** may determine that the short circuit occurs when the voltage across both ends of the second switch unit **112** is maintained under the preset voltage for a preset time. For example, the control unit **130** may determine that the short circuit occurs when the voltage across both ends of the second switch unit **112** is maintained under 10 [V] for 0.1 s. The control unit **130** may calculate a value of the voltage across both ends of the second switch unit **112** using the voltage of the third end **Nc** sensed by the sensing unit **150**, and may determine the short circuit occurs by comparing the value of the voltage across both ends of the second switch unit **112** with the preset voltage value. Alternatively, the control unit **130** may estimate the voltage across both ends of the second switch unit **112** using a voltage across both ends of the battery **200**. In this case, the control unit **130** may estimate the voltage across both ends of the second switch unit **112** using the voltage of the second end **Nb** sensed by the sensing unit **150**.

Further, the control unit **130** may determine whether the disconnection occurs using the voltage across both ends of the inverter, which is measured when the second switch unit **112** is turned on and the first switch unit **111** is turned off. The control unit **130** may determine whether the disconnection occurs after a third set time has elapsed after operating the second switch unit **112** in the on state, and turning off the first switch unit **111**. In this case, the control unit **130** may determine whether the disconnection occurs after making

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the DC/DC converter **100** enter the charging mode. Specifically, the control unit **130** may determine whether the disconnection occurs by comparing a preset third reference voltage value and a preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of the inverter after the third set time has elapsed. The third set time may be set to a predetermined time after the second switch unit **112** is converted to an on state and the first switch unit **111** is converted to an off state. For example, the third set time may be set to 10.0 ms, but is not limited thereto. The control unit **130** may calculate the third inverter voltage value and the third inverter current value using the voltage and the current of the first end Na sensed by the sensing unit **150**, and may determine whether the disconnection occurs by comparing the third inverter voltage value and the third inverter current value with the preset third reference voltage value and the preset third reference current value.

Further, when it is determined that the disconnection occurs in a state of the charging mode, the control unit **130** may enter the discharging mode and reduce the inverter voltage. The control unit **130** may determine whether the disconnection additionally occurs in a state in which the voltage of the inverter end is reduced. The control unit **130** may determine whether the disconnection occurs by comparing the preset third reference voltage value and the preset third reference current value with the third inverter voltage value and the third inverter current value measured at both ends of the inverter in a state of entering the discharging mode. The control unit **130** may block introduction of the current even after entering the discharge mode by operating the circuit breaker of the top switch unit **110** when it is determined that the disconnection continuously occurs.

FIG. 5 is a flow chart of an operation of the DC/DC converter according to the embodiment.

According to the embodiment, first, a voltage difference between a battery and an inverter may be reduced and an inrush current may be blocked by initially charging (or pre-charging) a direct current link capacitor disposed at an input end of the inverter (**S501**).

Next, a control unit may turn on a first switch unit located between the direct current link capacitor and a bridge circuit unit (**S502**).

Next, a sensing unit may measure the voltage and current across both ends of the inverter after a first set time has elapsed (**S503**).

Next, the control unit determine whether a short circuit occurs by comparing a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of the inverter (**S504**).

When it is determined that the short circuit occurs, the control unit may block the introduction of the current by operating a top switch unit (**S505**).

Next, the sensing unit may measure the voltage and current across both ends of the inverter after a second set time has elapsed (**S506**).

Next, the control unit may determine whether the short circuit and the disconnection occur by comparing a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of the inverter (**S507**).

When it is determined that the disconnection or short circuit occurs, the control unit may block the introduction of the current by operating the top switch unit (**S508**).

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Next, the sensing unit may measure the voltage across both ends of a second switch unit connected in parallel with the first switch unit (**S509**).

Next, the control unit may determine whether the short circuit occurs by comparing the voltage across both ends of the second switch unit and a preset voltage. The control unit may determine that the short circuit occurs when the voltage across both ends of the second switch unit is maintained under the preset voltage for a preset time (**S510**).

When it is determined that the short circuit occurs, the control unit may block the introduction of the current by operating the top switch unit (**S511**).

Next, the control unit may turn on the second switch unit (**S512**).

Next, the control unit may turn off the first switch unit. Accordingly, the converter may enter a charging mode (**S513**).

Next, the sensing unit may measure the voltage and current across both ends of the inverter after a third set time has elapsed (**S514**).

Next, the control unit may determine whether the disconnection occurs by comparing a preset third reference voltage value and a preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of the inverter (**S515**).

Next, when it is determined that the disconnection occurs in the charging mode state, the control unit may enter a discharging mode and reduce the voltage of the inverter end (**S516**).

Next, the sensing unit may measure the voltage and current across both ends of the inverter (**S517**).

Next, the control unit may determine whether the disconnection additionally occurs in the discharging mode. The control unit may determine whether the disconnection occurs by comparing the comparing the preset third reference voltage value and the preset third reference current value with the third inverter voltage value and the third inverter current value measured at both ends of the inverter, and then checking whether the voltage is reduced in the state of entering the discharging mode (**S518** and **S519**).

The control unit may block the introduction of the current by operating the top switch unit **110** when it is determined that the disconnection continuously occurs even after entering the discharge mode (**S520**).

FIGS. 6A-6D are views of a result of testing detection performance of the short circuit and the disconnection of the DC/DC converter according to the embodiment. Referring to FIG. 6A, since an existing DC/DC converter performs initial charging without determining whether a short circuit and a disconnection occur, it can be seen that an error PreChargeERR according to the short circuit or disconnection occurs during the initial charging. In comparison of this, referring to FIG. 6B, the DC/DC converter according to the embodiment may determine whether two short circuits occur (Short Circuit 0 and 1) and whether one disconnection occurs (OTD1) in a state in which the first switch unit is initially turned on. Further, referring to FIG. 6C, by determining whether an additional disconnection occurs (OTD2) in a state in which the first switch unit is converted to the off state and the second switch unit is turned on, as shown in FIG. 6D, there is a technical effect in that a stable initial operation can be performed.

The term ‘-unit’ used in the embodiment refers to software or a hardware component such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC), and the ‘-unit’ performs certain roles. However, the ‘-unit’ is not a meaning limited to the software or hardware.

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The ‘-unit’ may be configured to be present in an addressable storage medium, and may be configured to play one or more processors. Accordingly, for example, the ‘-unit’ includes components such as software components, object oriented software components, class components, and task components, processes, functions, properties, procedures, subroutines, segments of program code, drivers, firmware, micro-code, circuits, data, databases, data structures, tables, arrays, and variables. Functions provided in the components and the ‘-units’ may be combined into a smaller number of components and ‘-units’ or further separated into additional components and ‘-units’. Further, the components and the ‘-units’ may be implemented to play one or more central processing units (CPUs) in a device or a security multimedia card.

Although the preferable embodiments of the present invention are described above, those skilled in the art may variously modify and change the present invention within the scope and spirit of the present invention disclosed in the following claims.

The invention claimed is:

1. A direct current/direct current (DC/DC) converter control method comprising:

turning on a first switch unit located between a direct current link capacitor and a bridge circuit unit;

determining whether a short circuit (short) occurs after a first set time has elapsed;

determining whether the short circuit and a disconnection (open) occur after a second set time has elapsed;

measuring a voltage across both ends of a second switch unit connected in parallel with the first switch unit;

comparing the voltage across both ends of the second switch unit and a preset voltage to determine whether the short circuit occurs;

turning on the second switch unit;

turning off the first switch unit; and

determining whether the disconnection occurs after a third set time has elapsed from a time when the first switch unit is turned off.

2. The DC/DC converter control method of claim 1, wherein the determining of whether the short circuit (short) occurs after the first set time has elapsed includes comparing a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of an inverter and determining whether the short circuit occurs.

3. The DC/DC converter control method of claim 1, wherein the determining of whether the short circuit and the disconnection (open) occur after the second set time has elapsed includes comparing a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of an inverter and determining whether the short circuit and the disconnection occur.

4. The DC/DC converter control method of claim 1, wherein the determining of whether the disconnection occurs after the third set time has elapsed includes comparing a preset third reference voltage value and a preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of an inverter and determining whether the short circuit and the disconnection occur.

5. The DC/DC converter control method of claim 1, wherein the comparing of the voltage across both ends of the second switch unit and the preset voltage to determine whether the short circuit occurs includes determining that

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the short circuit occurs when the voltage across both ends of the second switch unit is maintained under the preset voltage for a preset time.

6. The DC/DC converter control method of claim 1, comprising outputting at least one of a control command for stopping an operation of a converter and a control command for reducing a voltage of an inverter end when the short circuit or the disconnection occurs.

7. A direct current/direct current (DC/DC) converter comprising:

a first switch unit located between a direct current link capacitor and a bridge circuit unit;

a second switch unit connected in parallel with the first switch unit;

a sensing unit configured to sense a voltage across both ends of an inverter and a voltage across both ends of the second switch unit; and

a control unit configured to control an on-off operation of the first switch unit and the second switch unit, and determine whether a short circuit and a disconnection occur using the voltage across both ends of the inverter and the voltage across both ends of the second switch unit,

wherein the control unit determines whether the short circuit and the disconnection occur using the voltage across both ends of the inverter and the voltage across both ends of the second switch unit, which are measured when the first switch unit is turned on and the second switch unit is turned off; and determines whether the disconnection occurs using the voltage across both ends of the inverter, which is measured when the second switch unit is turned on and the first switch unit is turned off.

8. The DC/DC converter of claim 7, wherein the first switch unit includes a resistor element and a switching element connected to the resistor element in series.

9. The DC/DC converter of claim 7, wherein the control unit sequentially performs arithmetic operations of determining whether a short circuit (short) occurs after a first set time has elapsed after turning on the first switch unit, determining whether the short circuit and a disconnection (open) occur after a second set time has elapsed, and comparing the voltage across both ends of the second switch unit and a preset voltage to determine whether the short circuit occurs, and determines whether the disconnection occurs after a third set time has elapsed after turning on the second switch unit and turning off the first switch unit.

10. The DC/DC converter of claim 9, wherein the control unit outputs at least one of a control command for stopping an operation of a converter and a control command for reducing a voltage of an inverter end when the short circuit or the disconnection occurs.

11. The DC/DC converter of claim 9, wherein the control unit compares a preset first reference voltage value and a preset first reference current value with a first inverter voltage value and a first inverter current value measured at both ends of the inverter after a first set time has elapsed to determine whether the short circuit occurs.

12. The DC/DC converter of claim 9, wherein the control unit compares a preset second reference voltage value and a preset second reference current value with a second inverter voltage value and a second inverter current value measured at both ends of the inverter after the second set time has elapsed to determine whether the short circuit and the disconnection occur.

13. The DC/DC converter of claim 9, wherein the control unit compares a preset third reference voltage value and a

preset third reference current value with a third inverter voltage value and a third inverter current value measured at both ends of the inverter after the third set time has elapsed to determine whether the short circuit and the disconnection occur.

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14. The DC/DC converter control method of claim 6, comprising blocking introduction of a current by operating a circuit breaker of a top switch unit including the first switch unit and the second switch unit when the short circuit or the disconnection occurs.

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15. The DC/DC converter control method of claim 1, wherein the second set time is set to be longer than the first set time.

16. The DC/DC converter of claim 7, wherein the control unit determines that the short circuit occurs when the voltage across both ends of the second switch unit is maintained under a preset voltage for a preset time.

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17. The DC/DC converter of claim 7, wherein the control unit comprises blocking introduction of a current by operating a circuit breaker of a top switch unit including the first switch unit and the second switch unit when the short circuit or the disconnection occurs.

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18. The DC/DC converter of claim 9, wherein the second set time is set to be longer than the first set time.

19. The DC/DC converter of claim 7, wherein the first switch unit includes a first switching element and a resistor.

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20. The DC/DC converter of claim 19, wherein the first switch element has one side connected to one end of the direct current link capacitor and an other side connected to one side of the resistor, and the resistor has one side connected to the other side of the first switch element and an other side connected to the bridge circuit unit.

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